# USB camera preview

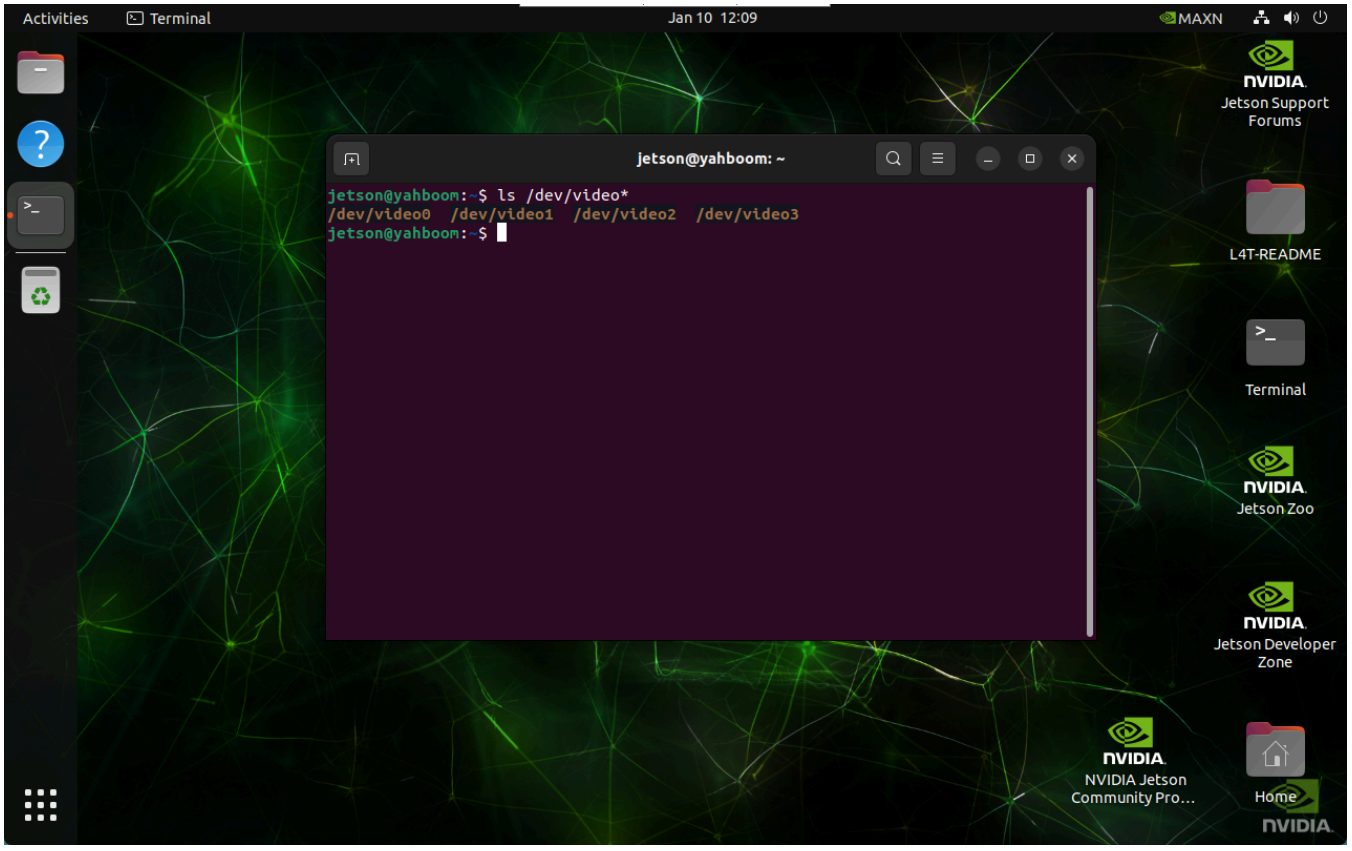
**USB camera preview**

1. View the video devices
2. GUVCView
    2.1. GUVCView installation
    2.2. GUVCView usage
3. VLC
    3.1. VLC installation
    3.2. VLC usage

## 1. View the video devices

```
ls /dev/video*
```

The result shown in the image is the result of connecting two CSI cameras and one USB camera: generally a CSI camera shows one `video` device, while a USB camera shows two `video` devices. For the USB camera, select the newly added `/dev/video2` with the smaller number (connecting the USB camera adds the `/dev/video2` and `/dev/video3` device numbers to the system).

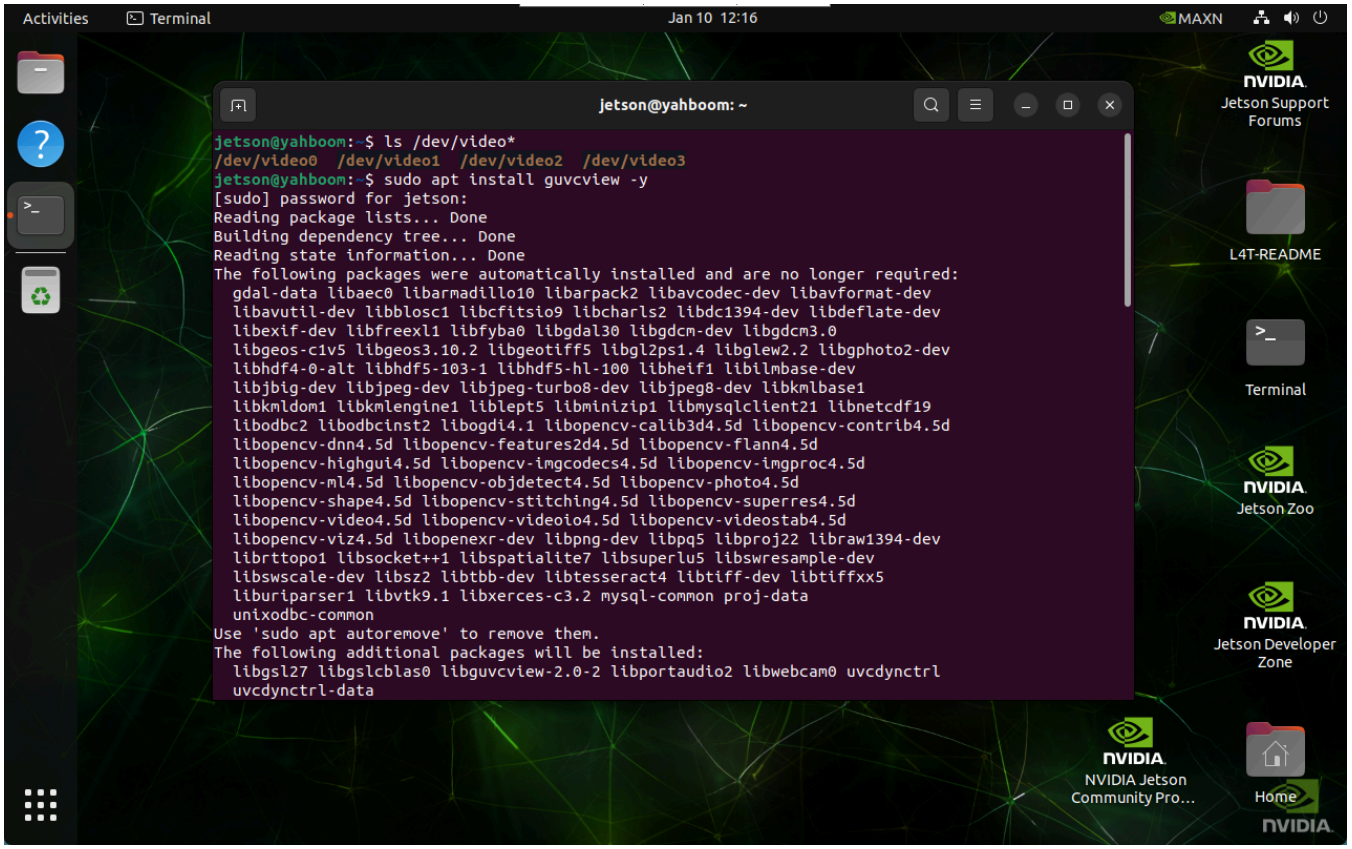



## 2. GUVCView

GUVCView is open source software for Linux systems used to capture and record video and images, mainly for Webcam cameras.

## 2.1. GUVCView installation

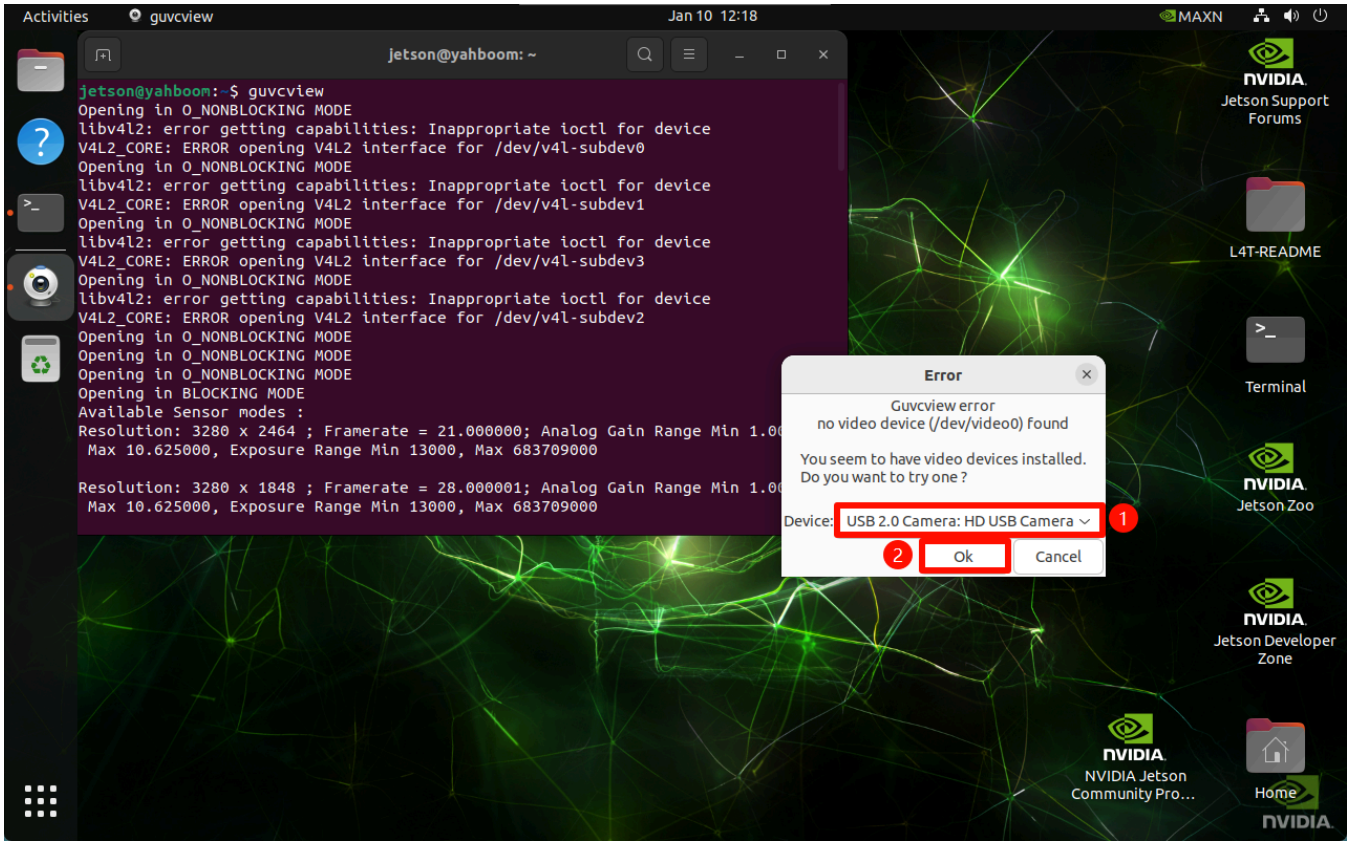
```
sudo apt update
sudo apt install guvcview -y
```

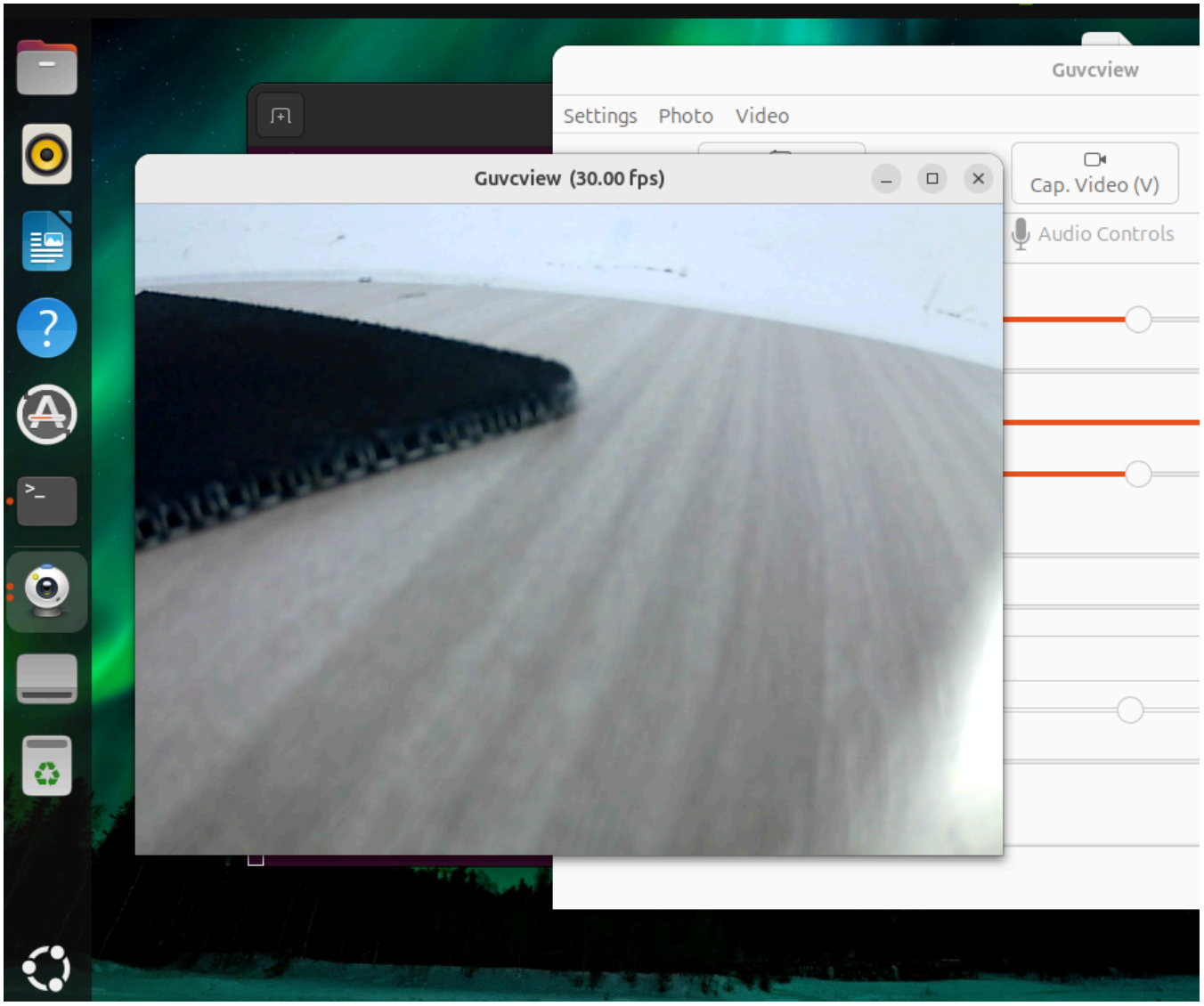



## 2.2. GUVCView usage

Enter the application menu bar and click the `guvcview` icon, or enter the startup command in the terminal: select the USB camera; the CSI camera shows no preview image.

```
guvcview
```


Activities
guvcview
Jan 10 12:18
MAXN
jetson@yahboom: ~
jetson@yahboom:~$ guvcview
Opening in O_NONBLOCKING MODE
libv4l2: error getting capabilities: Inappropriate ioctl for device
V4L2_CORE: ERROR opening V4L2 interface for /dev/v4l-subdev0
Opening in O_NONBLOCKING MODE
libv4l2: error getting capabilities: Inappropriate ioctl for device
V4L2_CORE: ERROR opening V4L2 interface for /dev/v4l-subdev1
Opening in O_NONBLOCKING MODE
libv4l2: error getting capabilities: Inappropriate ioctl for device
V4L2_CORE: ERROR opening V4L2 interface for /dev/v4l-subdev3
Opening in O_NONBLOCKING MODE
libv4l2: error getting capabilities: Inappropriate ioctl for device
V4L2_CORE: ERROR opening V4L2 interface for /dev/v4l-subdev2
Opening in O_NONBLOCKING MODE
Opening in O_NONBLOCKING MODE
Opening in O_NONBLOCKING MODE
Opening in BLOCKING MODE
Available Sensor modes :
Resolution: 3280 x 2464 ; Framerate = 21.000000; Analog Gain Range Min 1.0
Max 10.625000, Exposure Range Min 13000, Max 683709000
Resolution: 3280 x 1848 ; Framerate = 28.000001; Analog Gain Range Min 1.0
Max 10.625000, Exposure Range Min 13000, Max 683709000
Error
Guvcview error
no video device (/dev/video0) found
You seem to have video devices installed.
Do you want to try one ?
Device:
USB 2.0 Camera: HD USB Camera
1
2
Ok
Cancel
NVIDIA
Jetson Support Forums
L4T-README
Terminal
Jetson Zoo
Jetson Developer Zone
NVIDIA Jetson Community Pro...
Home

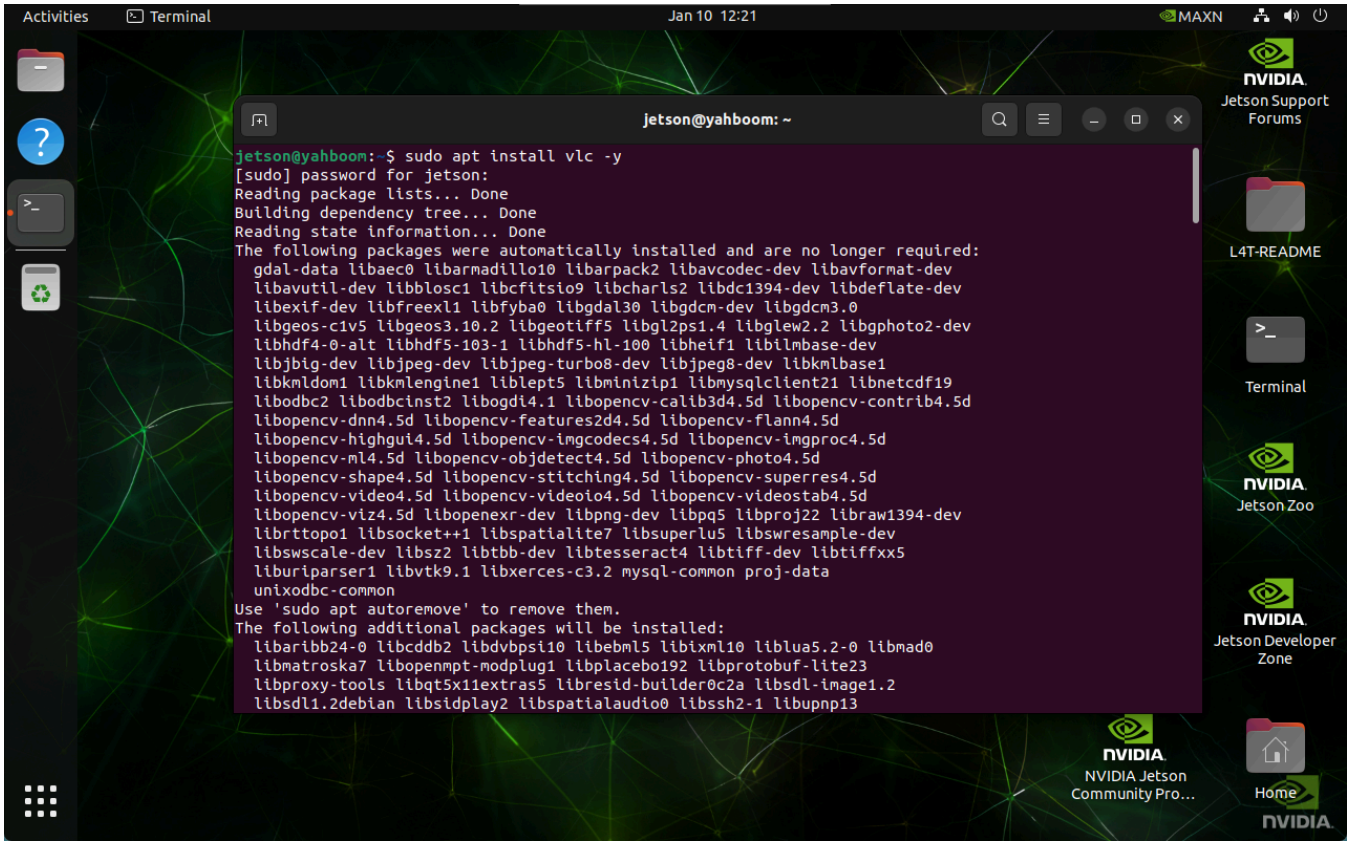



# 3. VLC

VLC media player is a free and open source multimedia player that supports many audio and video formats as well as DVD, audio CD, VCD and various streaming protocols.

## 3.1. VLC installation

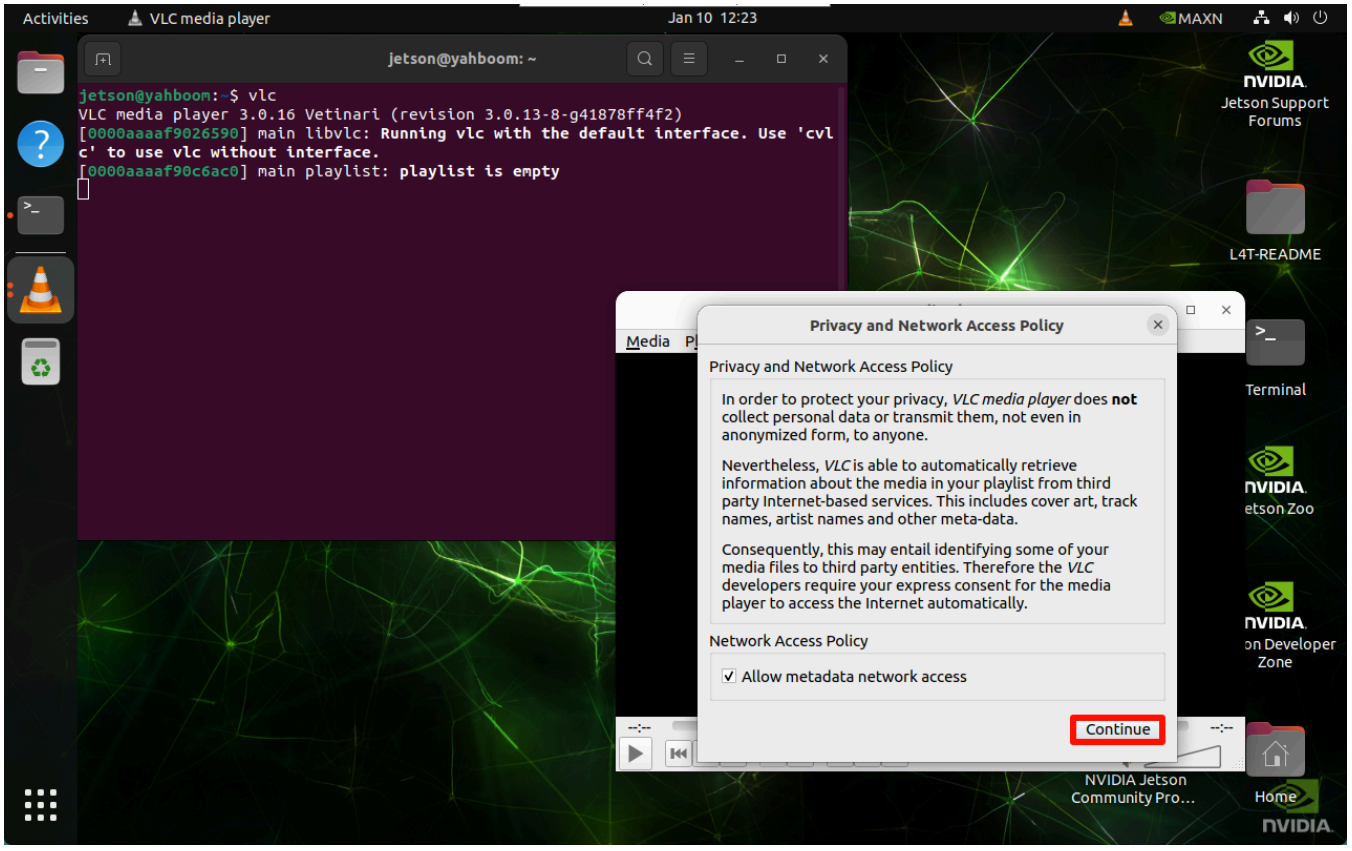
```
sudo apt update
sudo apt install vlc -y
```

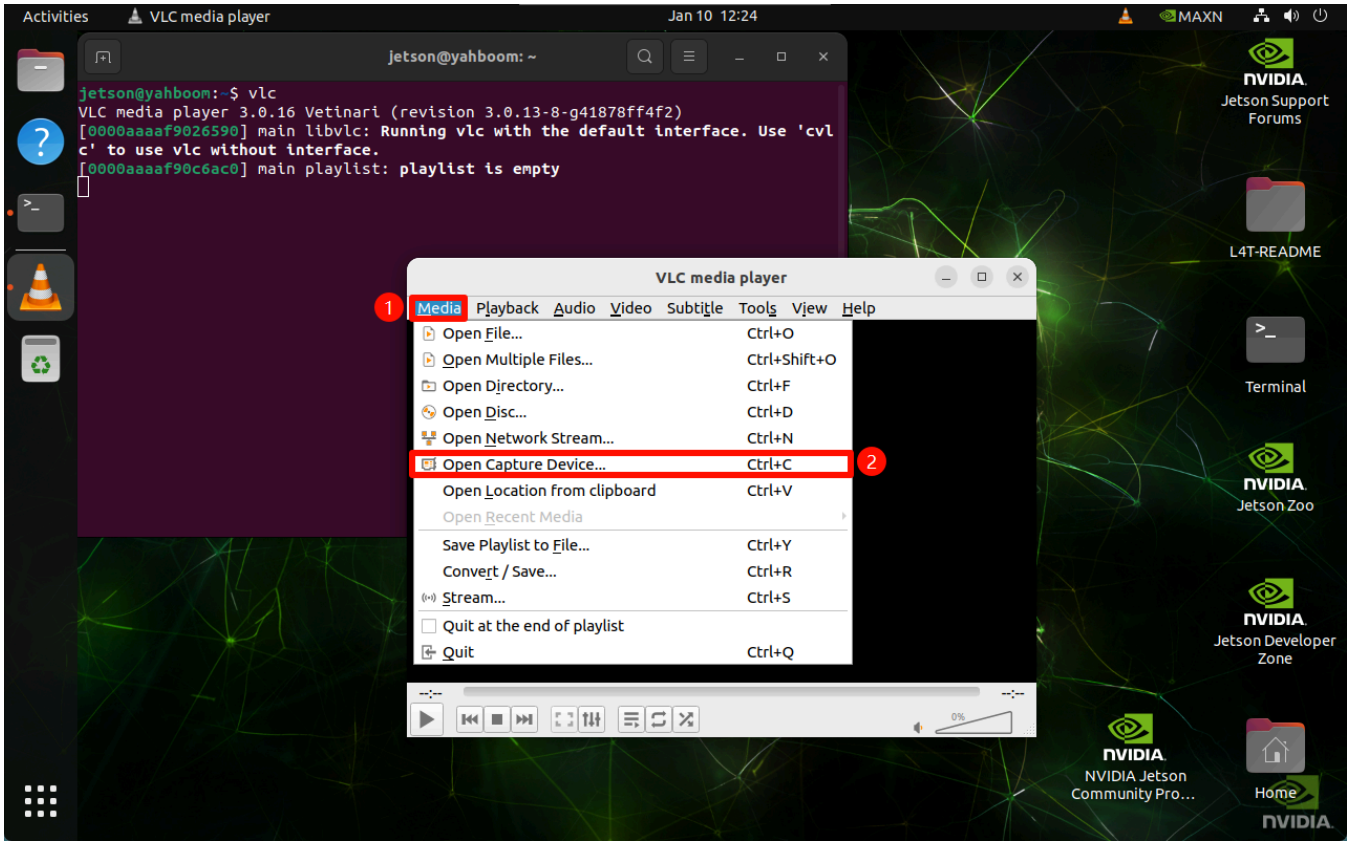



## 3.2. VLC usage

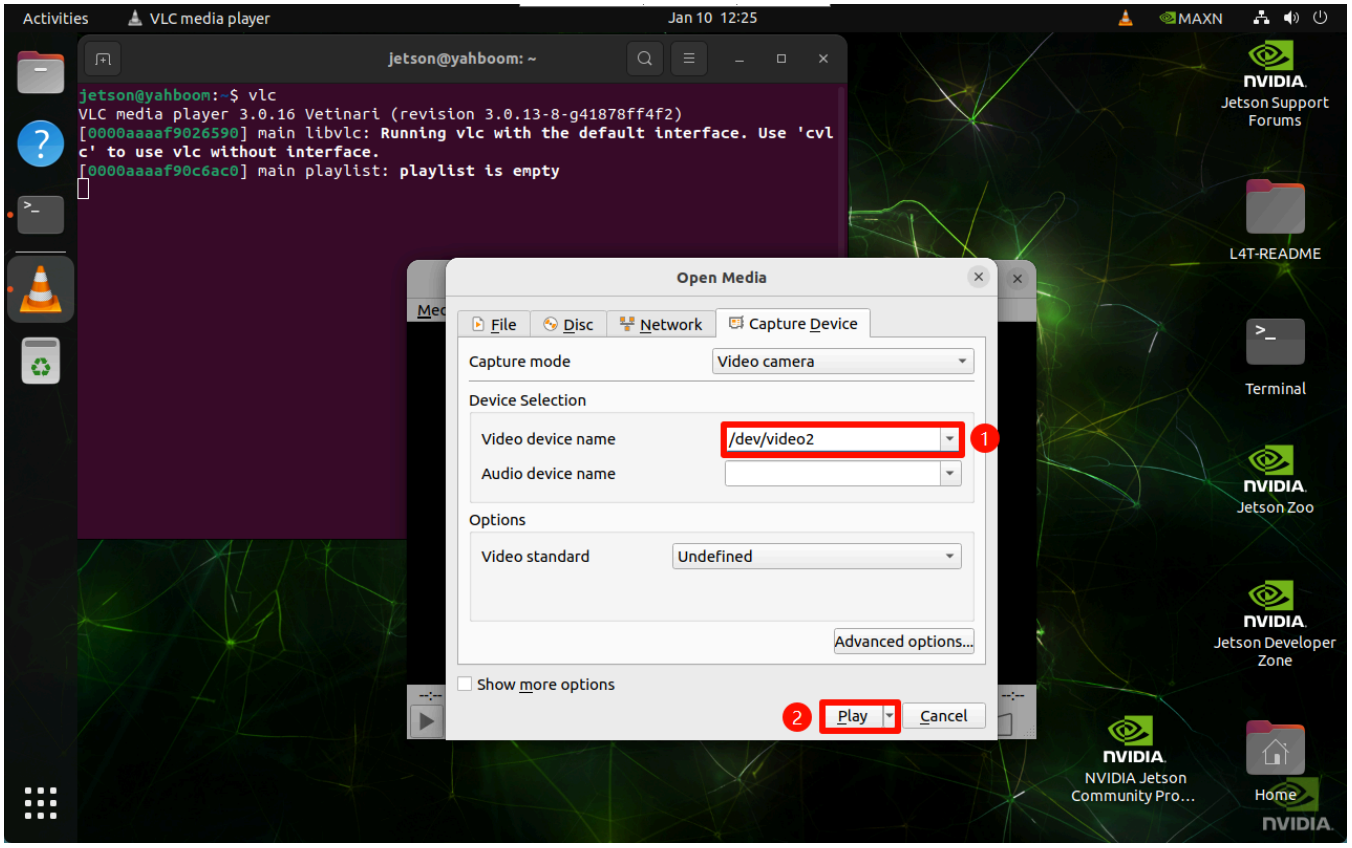
Enter the application menu bar and click the `VLC media player` icon, or enter the startup command in the terminal: select the USB camera; the CSI camera shows no preview image.

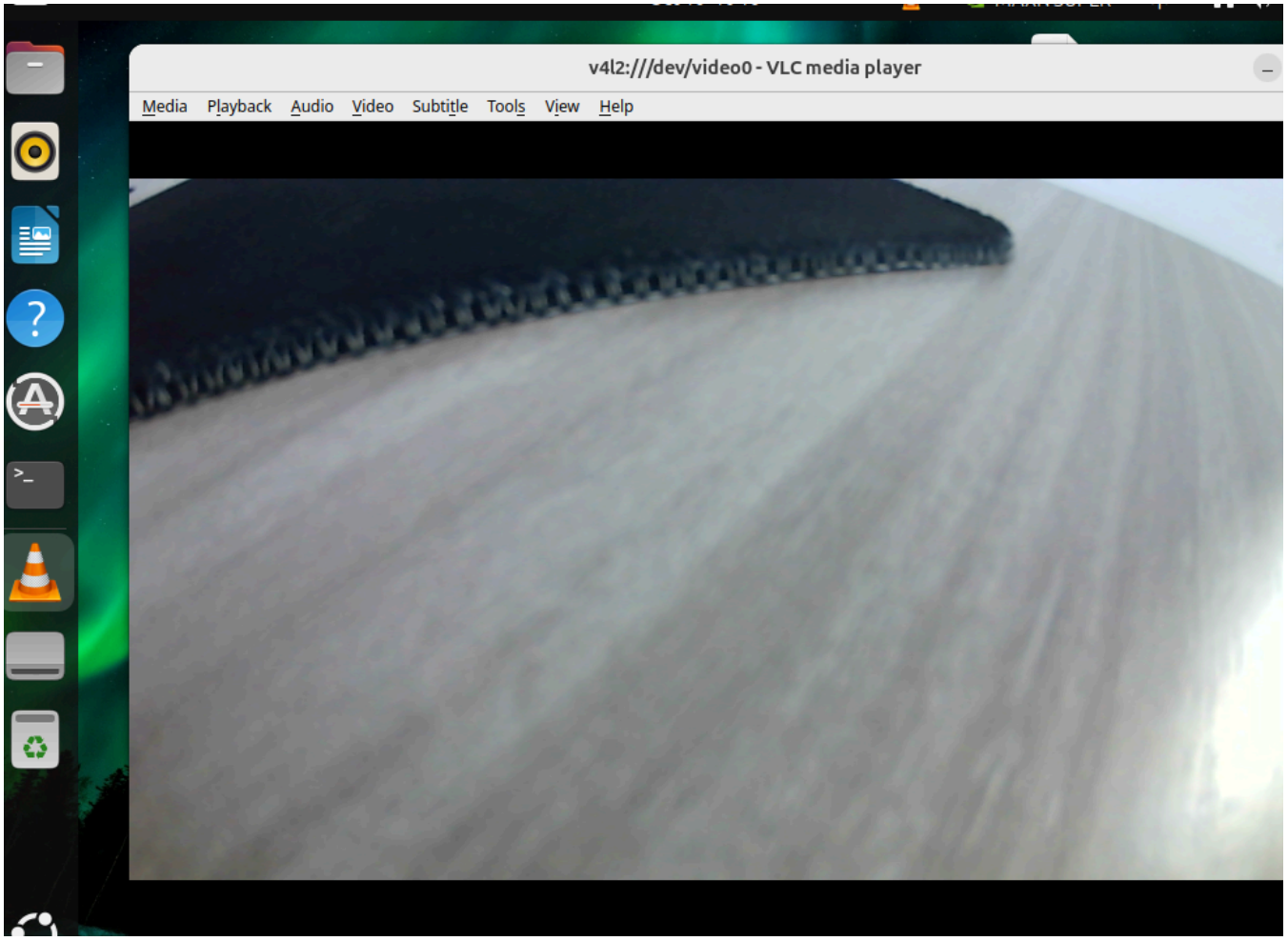
```
vlc
```






Select the device number that corresponds to the USB camera:




v4l2:///dev/video0 - VLC media player
Media
Playback
Audio
Video
Subtitle
Tools
View
Help